

Qtangl Q-Day Readiness Report

Scan ID: scan-fe310fb9-4900-4136-a734-10bb285be6b7

Target: api.regionalbank.example

Readiness band: Pre-migration baseline

Readiness score: 15.5/100

Generated: 2026-06-21

Scan depth: standard

Migration exposure: \$175,000–\$595,000 (order-of-magnitude migration exposure)

Nearest deadline: 2025-ongoing

0.0% of 7 discovered endpoints negotiate hybrid/PQC today. 7 asset(s) require migration under NIST IR 8547 timelines. Coverage is endpoint-scoped, not a formal audit.

Peer benchmark: enroll Monitor for industry comparison when cohort size allows.

Scope and authorization

Field	Value
Authorized target	api.regionalbank.example
Scan depth	standard
Scenario	bank-tls-inventory
Classified assets	7
Surfaces observed	code_signing, discovery, email, jwks, ssh, tls
Authorization recorded	—

Explicitly out of scope

- Internal HSM and offline key material
- Shadow APIs not reachable from scan target
- Application-layer cryptography not exposed on scanned ports
- This scan is an inventory aid, not a formal cryptographic audit or penetration test.
- Quantum-vulnerable does not mean broken today; RSA/ECC remain classically secure until cryptographically relevant QC.

Findings delta vs manual baseline

Qtangl classified 7 asset(s) vs manual baseline of 42.

Metric	Manual baseline	Qtangl scan
Wall time	8.0 weeks	0.0s
Assets discovered	42	7
Quantum vulnerable	38	7
HNDL exposed	—	7
Net-new assets	—	0

Customer attestation (optional): We confirm net-new findings vs our prior inventory dated _____.

Manual vs Qtangl scoreboard

Metric	Customer baseline	Qtangl discovery
Wall time	8.0 weeks	0.0s
Assets	42	7
Quantum vulnerable	38	7
HNDL exposed	38	7
Readiness	18.0	15.5

Readiness score	15.5/100
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Readiness band	Pre-migration baseline
Coverage confidence	68.0%
Crypto agility score	23.6
Assets classified	7
Remediation items	7

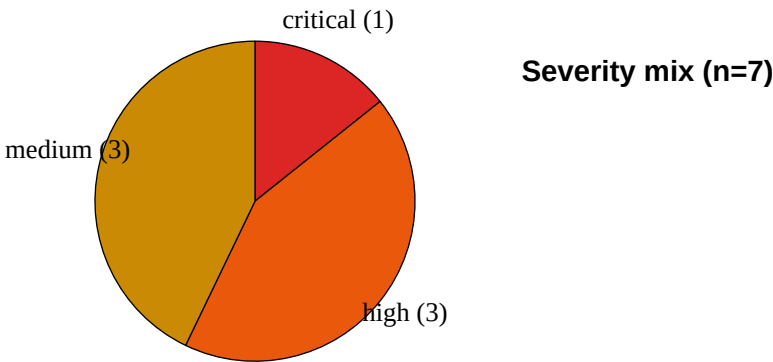
This endpoint-scoped Q-Day readiness assessment identified **7** classified cryptographic assets against **api.regionalbank.example**. **7** require migration under current NIST timelines; **0** already negotiate hybrid/PQC.

Priority: Migrate Release artifact code signing — Release artifact code signing uses RSA which an adversary could break with a future quantum computer. Intercepted traffic today may be decrypted later (HNDL risk).

Priority: Migrate OIDC JWKS RS256 — OIDC JWKS RS256 uses RS256 which an adversary could break with a future quantum computer. Intercepted traffic today may be decrypted later (HNDL risk).

Priority: Migrate API gateway TLS — API gateway TLS uses RSA which an adversary could break with a future quantum computer. Intercepted traffic today may be decrypted later (HNDL risk).

Risk dashboard



Risk quadrant (severity × HNDL)



Algorithm families (n=7)



Mosca inequality (HNDL)

Mosca inequality holds: X+Y (15.0 yr) > Z (12.0 yr). Sensitive data may be decrypted after Q-Day if not migrated in time.

Mosca timeline (years)



Under these assumptions, ciphertext captured today may be decrypted after Q-Day unless mig

Cryptographic inventory

Asset	Algorithm	Severity	PQC	Shor qubits
Release artifact code signing	RSA	critical	No	8194
OIDC JWKS RS256	RS256	high	No	4098
API gateway TLS	RSA	high	No	4098
Customer auth TLS	ECDSA	high	No	2330
Ops bastion SSH host key	ssh-rsa	medium	No	6146
Inbound SMTP STARTTLS	RSA	medium	No	—
CT-discovered subdomain	ECDSA	medium	No	—

Release artifact code signing — What this means: Release artifact code signing uses RSA which an adversary could break with a future quantum computer. Intercepted traffic today may be decrypted later (HNDL risk).

OIDC JWKS RS256 — What this means: OIDC JWKS RS256 uses RS256 which an adversary could break with a future quantum computer. Intercepted traffic today may be decrypted later (HNDL risk).

API gateway TLS — What this means: API gateway TLS uses RSA which an adversary could break with a future quantum computer. Intercepted traffic today may be decrypted later (HNDL risk).

Customer auth TLS — What this means: Customer auth TLS uses ECDSA which an adversary could break with a future quantum computer. Intercepted traffic today may be decrypted later (HNDL risk).

Ops bastion SSH host key — What this means: Ops bastion SSH host key uses ssh-rsa which an adversary could break with a future quantum computer. Intercepted traffic today may be decrypted later (HNDL risk).

Inbound SMTP STARTTLS — What this means: Inbound SMTP STARTTLS uses RSA which an adversary could break with a future quantum computer. Intercepted traffic today may be decrypted later (HNDL risk).

CT-discovered subdomain — What this means: CT-discovered subdomain uses ECDSA which an adversary could break with a future quantum computer. Intercepted traffic today may be decrypted later (HNDL risk).

Top HNDL-exposed assets (deep dive)

Release artifact code signing [ALREADY TOO LATE] — priority 245

TLS — | Cipher: — | Group: —

What this means: Release artifact code signing uses RSA which an adversary could break with a future quantum computer. Intercepted traffic today may be decrypted later (HNDL risk).

OIDC JWKS RS256 [ALREADY TOO LATE] — priority 158

TLS — | Cipher: — | Group: —

What this means: OIDC JWKS RS256 uses RS256 which an adversary could break with a future quantum computer. Intercepted traffic today may be decrypted later (HNDL risk).

API gateway TLS [ALREADY TOO LATE] — priority 144

TLS TLSv1.3 | Cipher: TLS_AES_256_GCM_SHA384 | Group: x25519

What this means: API gateway TLS uses RSA which an adversary could break with a future quantum computer. Intercepted traffic today may be decrypted later (HNDL risk).

Customer auth TLS [ALREADY TOO LATE] — priority 144

TLS TLSv1.3 | Cipher: TLS_AES_128_GCM_SHA256 | Group: secp256r1

What this means: Customer auth TLS uses ECDSA which an adversary could break with a future quantum computer. Intercepted traffic today may be decrypted later (HNDL risk).

Ops bastion SSH host key [ALREADY TOO LATE] — priority 101

TLS — | Cipher: — | Group: —

What this means: Ops bastion SSH host key uses ssh-rsa which an adversary could break with a future quantum computer. Intercepted traffic today may be decrypted later (HNDL risk).

Prioritized remediation backlog

#	Title	Severity	PQC algorithm	Deadline	Effort
1	Migrate Release artifact code signing	critical	Migrate code/document signing to SL	2030	90d
2	Migrate OIDC JWKS RS256	high	Rotate JWT signing keys to ML-DSA-6	Available now (2024)	30d
3	Migrate API gateway TLS	high	Migrate to ML-KEM-768 hybrid key ex	2030	45d
4	Migrate Customer auth TLS	high	Deploy TLS 1.3 hybrid KEX	Available now (2024)	45d
5	Migrate Ops bastion SSH host key	medium	Replace SSH RSA host keys with ML-D	2030	21d
6	Migrate Inbound SMTP STARTTLS	medium	Enable STARTTLS with ML-KEM hybrid	2030	60d
7	Migrate CT-discovered subdomain	medium	ML-DSA-65	Available now (2024)	30d

Migration roadmap

Milestone	Deadline	Severity	PQC / notes
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Prioritize 7 HNDL-exposed asset(s)	Immediate	critical	
Migrate Release artifact code signing	2030	critical	Migrate code/document signing
Migrate OIDC JWKS RS256	Available now (2024)	high	Rotate JWT signing keys to ML-
Migrate API gateway TLS	2030	high	Migrate to ML-KEM-768 hybrid k
Migrate Customer auth TLS	Available now (2024)	high	Deploy TLS 1.3 hybrid KEX
Migrate Ops bastion SSH host key	2030	medium	Replace SSH RSA host keys with
Migrate Inbound SMTP STARTTLS	2030	medium	Enable STARTTLS with ML-KEM hy
Migrate CT-discovered subdomain	Available now (2024)	medium	ML-DSA-65
NIST IR 8547 transition	2030	info	nist-ir-8547
CNSA 2.0	2030-2033	info	cnsa-2.0
NSM-10	2035	info	nsm-10

Remediation what-if (estimate only)

Scenario	Items	Projected score	Delta
Top 3 backlog items	3	33.5	+18.0
All critical/high items	4	39.5	+24.0
Full backlog complete	7	45.5	+30.0

Maturity stage 0: Unaware — next: Inventory

Compliance framework mapping

Regional bank TLS inventory — compliance pack

PCI-DSS 4.0 [gap]: Cryptographic agility and key management requirements

NIST IR 8547 [gap]: Transitioning to post-quantum cryptography standards

CNSA 2.0 [gap]: Commercial National Security Algorithm Suite 2.0

NSM-10 [gap]: National Security Memorandum on post-quantum cryptography

FIPS 204 (ML-DSA) [gap]: Module-Lattice-Based Digital Signature Algorithm standard

CMMC / FedRAMP [gap]: Federal contractor cryptographic inventory expectations

Compliance summary: { 'controlsAtRisk': [{ 'framework': 'nist-ir-8547', 'ref': 'Migration planning', 'theme': 'Inventory quantum-vulnerable cryptography', 'status': 'at_risk', 'assetId': 'code-sign-legacy', 'assetLabel': 'Release artifact code signing' }, { 'framework': 'nist-ir-8547', 'ref': 'HNDL assessment', 'theme': 'Assess harvest-now-decrypt-later exposure', 'status': 'at_risk', 'assetId': 'code-sign-legacy', 'assetLabel': 'Release artifact code signing' }, { 'framework': 'pci-dss-4', 'ref': '4.2',

'theme': 'Protect cardholder data in transit', 'status': 'at_risk', 'assetId': 'code-sign-legacy', 'assetLabel': 'Release artifact code signing'}, {'framework': 'pci-dss-4', 'ref': '6.3', 'theme': 'Cryptographic agility and key lifecycle', 'status': 'at_risk', 'assetId': 'code-sign-legacy', 'assetLabel': 'Release artifact code signing'}, {'framework': 'pci-dss-4', 'ref': '4.2', 'theme': 'Protect cardholder data in transit', 'status': 'at_risk', 'assetId': 'jwks-oidc', 'assetLabel': 'OIDC JWKS RS256'}, {'framework': 'pci-dss-4', 'ref': '6.3', 'theme': 'Cryptographic agility and key lifecycle', 'status': 'at_risk', 'assetId': 'jwks-oidc', 'assetLabel': 'OIDC JWKS RS256'}, {'framework': 'nist-ir-8547', 'ref': 'Migration planning', 'theme': 'Inventory quantum-vulnerable cryptography', 'status': 'at_risk', 'assetId': 'tls-api-bank', 'assetLabel': 'API gateway TLS'}, {'framework': 'nist-ir-8547', 'ref': 'HNDL assessment', 'theme': 'Assess harvest-now-decrypt-later exposure', 'status': 'at_risk', 'assetId': 'tls-api-bank', 'assetLabel': 'API gateway TLS'}, {'framework': 'pci-dss-4', 'ref': '4.2', 'theme': 'Protect cardholder data in transit', 'status': 'at_risk', 'assetId': 'tls-api-bank', 'assetLabel': 'API gateway TLS'}, {'framework': 'pci-dss-4', 'ref': '6.3', 'theme': 'Cryptographic agility and key lifecycle', 'status': 'at_risk', 'assetId': 'tls-api-bank', 'assetLabel': 'API gateway TLS'}, {'framework': 'nist-ir-8547', 'ref': 'Migration planning', 'theme': 'Inventory quantum-vulnerable cryptography', 'status': 'at_risk', 'assetId': 'tls-auth-bank', 'assetLabel': 'Customer auth TLS'}, {'framework': 'nist-ir-8547', 'ref': 'HNDL assessment', 'theme': 'Assess harvest-now-decrypt-later exposure', 'status': 'at_risk', 'assetId': 'tls-auth-bank', 'assetLabel': 'Customer auth TLS'}, {'framework': 'pci-dss-4', 'ref': '4.2', 'theme': 'Protect cardholder data in transit', 'status': 'at_risk', 'assetId': 'tls-auth-bank', 'assetLabel': 'Customer auth TLS'}, {'framework': 'pci-dss-4', 'ref': '6.3', 'theme': 'Cryptographic agility and key lifecycle', 'status': 'at_risk', 'assetId': 'tls-auth-bank', 'assetLabel': 'Customer auth TLS'}, {'framework': 'nist-ir-8547', 'ref': 'Migration planning', 'theme': 'Inventory quantum-vulnerable cryptography', 'status': 'at_risk', 'assetId': 'ssh-bastion', 'assetLabel': 'Ops bastion SSH host key'}, {'framework': 'nist-ir-8547', 'ref': 'HNDL assessment', 'theme': 'Assess harvest-now-decrypt-later exposure', 'status': 'at_risk', 'assetId': 'ssh-bastion', 'assetLabel': 'Ops bastion SSH host key'}, {'framework': 'pci-dss-4', 'ref': '4.2', 'theme': 'Protect cardholder data in transit', 'status': 'at_risk', 'assetId': 'ssh-bastion', 'assetLabel': 'Ops bastion SSH host key'}, {'framework': 'pci-dss-4', 'ref': '6.3', 'theme': 'Cryptographic agility and key lifecycle', 'status': 'at_risk', 'assetId': 'ssh-bastion', 'assetLabel': 'Ops bastion SSH host key'}, {'framework': 'nist-ir-8547', 'ref': 'Migration planning', 'theme': 'Inventory quantum-vulnerable cryptography', 'status': 'at_risk', 'assetId': 'email-mx', 'assetLabel': 'Inbound SMTP STARTTLS'}, {'framework': 'nist-ir-8547', 'ref': 'HNDL assessment', 'theme': 'Assess harvest-now-decrypt-later exposure', 'status': 'at_risk', 'assetId': 'email-mx', 'assetLabel': 'Inbound SMTP STARTTLS'}], 'controlsSatisfied': [], 'atRiskCount': 26, 'satisfiedCount': 0}

Discovery provenance

Source / surface	Assets
tls	2
code_signing	1
jwks	1
ssh	1
email	1
discovery	1

CBOM summary

CycloneDX 1.6 · schema qtangl-cbom-v1 · full export in cbom.json within the evidence bundle.

Component type	Count
tls	2
code_signing	1
jwks	1
ssh	1
email	1

discovery	1
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Import cbom.json into your GRC or procurement toolchain to track cryptographic components alongside software SBOMs.

Scoring methodology

Readiness = inventory_baseline + safe% – at_risk% – broken% + hybrid_credit (endpoint-scoped; not a formal audit).

This scan: baseline 27.5, hybrid credit 0.0, classified 7 → score 15.5/100 (Pre-migration baseline).

Bands: 0–25 Pre-migration baseline · 26–50 Partial · 51–75 In progress · 76–100 PQC-ready.

Crypto agility 23.6 — Moderate agility. PQC-ready 0.0% · short-lived certs 14.3%.

Shor logical-qubit estimates are order-of-magnitude references, not Q-Day timing predictions. See /assess/methodology and fixtures/calibration.md.

Methodology and honesty notes

- This scan is an inventory aid, not a formal cryptographic audit or penetration test.
- Quantum-vulnerable does not mean broken today; RSA/ECC remain classically secure until cryptographically relevant QC.
- Shor logical-qubit estimates are order-of-magnitude references, not predictions of Q-Day timing.
- Coverage is endpoint-scoped; shadow APIs, HSMs, and offline keys may be missed.

Report integrity & provenance

Algorithm: Ed25519

Key fingerprint: 841a4fce9b0542fd

Content hash (SHA-256): bbcd3f505f5bc5689db8d8c7cbe2f4bb0454242c264bc47123d801236a3b191a

Signed at: 2026-06-21T18:39:12.322881+00:00

Verify: <https://www.qtangl.com/verify?scanId=scan-fe310fb9-4900-4136-a734-10bb285be6b7>

